| INCH-POUND |

MIL-T-29504/5A
23 April 1990
SUPERSEDING
MIL-T-29504/5
25 August 1987

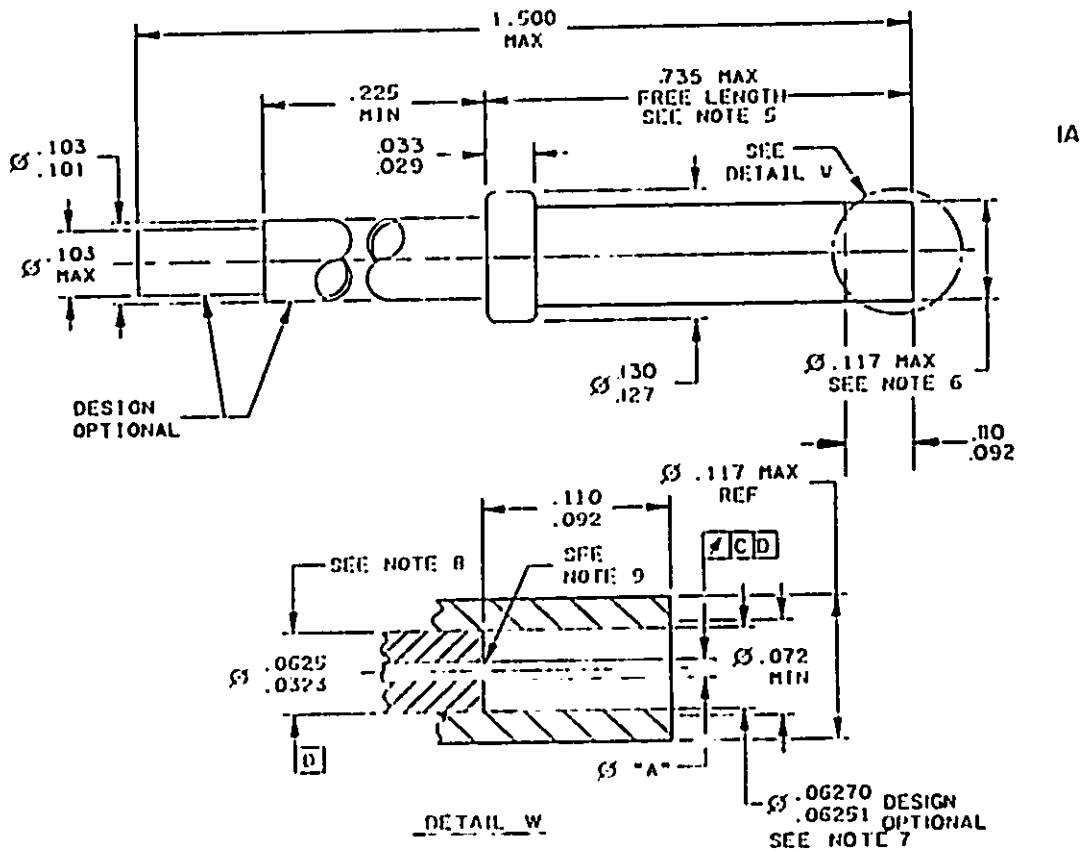
MILITARY SPECIFICATION SHEET

TERMINI, FIBER OPTIC, CONNECTOR, REMOVABLE, ENVIRONMENT RESISTING, CLASS 5, TYPE II, STYLE A, SOCKET TERMINUS, SIZE 16, REAR RELEASE MIL-C-38999, SERIES I, III, and IV

(A)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-29504.



IA

FIGURE 1. Socket terminus.

(A) denotes changes

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Inches	mm		Inches	mm
.029	0.74		.103	2.62
.033	0.84		.110	2.79
.072	1.83		.117	3.97
.0625	1.58		.127	3.23
.0627	1.59		.130	3.30
.084	2.13		.225	5.72
.092	2.34		.431	10.95
.100	2.54		.575	14.61
.101	2.57		.702	17.83
			1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within .003 unless otherwise indicated.
4. Dimensions apply after plating when applicable.
5. Free length prior to installation in the connector.
6. Alignment sleeve and body design optional, not to exceed maximum envelope dimensions.
7. For alternate alignment sleeve design, engaging force shall be less than preload force ((I.D.) tolerance does not apply).
8. Alignment surface O.D. shall be concentric with "A" dia per table I.
9. Polished fiber face may be flat to concave and flush to recess.
10. "A" diameter hole tolerance is ±8, see table I.

FIGURE 1. Socket terminus - Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Ⓐ Weight: 1 gram maximum.

Adhesives: Not applicable.

Tools: See table II.

Mating termini: MIL-T-29504/4.

Part or Identifying Number (PIN): See figure 1 and table I. The PIN M29504/5- (dash number from table I.)

Marking: The PIN shall be marked on barrel of the termini.

Terminus engagement and separation force: Minimum separation force shall be 2 ounces. Maximum average engagement force shall be 24 ounces. Maximum engagement force shall be 30 ounces.

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TABLE I. "A" diameter hole for optical waveguide fiber and PIN designation.

TICC	PIN	"A" diameter		Tolerance		Fiber size	Terminus
		µ	(inch) 1/	Circular runout 2/	Max $\frac{B + C}{2}$ 1/ (inch)µ	Core/clad µ	Size
4045	M29504/05-4045	124.5 +B -0	(.0049+B)	C			16
4046	M29504/05-4046	127.0 +B -0	(.0050+B)	C	(.00059) 15	50/125	16
4047	M29504/05-4047	129.5 +B -0	(.0051+B)	C			16
4048	M29504/05-4048	139.5 +B -0	(.0055+B)	C			16
4049	M29504/05-4049	142.0 +B -0	(.0056+B)	C	(.00059) 15	100/140	16
4050	M29504/05-4050	145.0 +B -0	(.0057+B)	C			16

1/ Dimensions in parentheses are inches.
2/ As defined by ANSI Y14.5M - 1982.

TABLE II. Tools.

End size	Insertion tool	Removal tool
16	M81969/14-03	M81969/14-03

Patent notice. The Government does not have a royalty-free license under the following listed patents for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Patent nunber	Patent expiration date
3,947,182	3/30/1993
4,351,586	9/28/1997
4,747,658	5/31/2005

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 11, 17, 19, 70, 80, 90, 99
DLA - ES

User activities:
Army - AV
Navy - CG, MC, OS, YD
Air Force - 13, 14, 84

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0078-5)

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